

DevOps Basic Interview Scripts

1. Shell Script - Service Monitoring

```
#!/bin/bash
SERVICE="nginx"

if systemctl is-active --quiet $SERVICE
then
    echo "$SERVICE is running"
else
    echo "$SERVICE is down. Restarting..."
    systemctl restart $SERVICE
fi
```

2. Terraform - EC2 Instance

```
provider "aws" {
    region = "us-east-1"
}

resource "aws_instance" "demo" {
    ami           = "ami-0c55b159cbfafa1f0"
    instance_type = "t2.micro"

    tags = {
        Name = "Interview-Instance"
    }
}
```

3. Ansible - Install Nginx

```
---
- name: Install Nginx
  hosts: all
  become: yes

  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
```

4. Jenkins Declarative Pipeline

```

pipeline {
  agent any

  stages {
    stage('Build') {
      steps {
        echo 'Building Application...'
      }
    }

    stage('Test') {
      steps {
        echo 'Running Tests...'
      }
    }

    stage('Deploy') {
      steps {
        echo 'Deploying Application...'
      }
    }
  }
}

```

5. Dockerfile - Node App

```

FROM node:18

WORKDIR /app

COPY package*.json ./
RUN npm install

COPY . .

EXPOSE 3000

CMD ["npm", "start"]

```

6. Kubernetes Deployment + Service

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: demo-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: demo
  template:
    metadata:
      labels:
        app: demo
    spec:
      containers:
        - name: demo-container
          image: nginx

```

```
        ports:
          - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: demo-service
spec:
  type: NodePort
  selector:
    app: demo
  ports:
    - port: 80
      targetPort: 80
```

7. Python - API Health Check

```
import requests

url = "https://example.com/health"
response = requests.get(url)

if response.status_code == 200:
    print("Application is Healthy")
else:
    print("Application is Down")
```